

UCS 503 - Software Engineering

MeterIQ

Explainable Smart Meter Analytics Platform

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Mid-Semester Project Evaluation

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Introduction

1.1 Background and Context

MeterIQ is an explainable smart-meter analytics platform designed to support the analysis of electricity consumption data. The system focuses on two major analytical tasks: identifying unusual consumption behaviour and performing short-term load forecasting.

The project uses smart-meter data from the Mathura and Bareilly regions. Smart-meter readings contain temporal consumption information that can be analysed to understand normal usage patterns, unusual observations, and future demand.

The platform is designed as a decision-support system for authorised users. Instead of automatically declaring an observation as electricity theft, the system is intended to identify anomalous behaviour and provide risk-oriented information that can support further human investigation.

The project combines data preparation, exploratory data analysis, feature engineering, machine-learning analysis, forecasting, and an interactive dashboard into a unified software system.

1.2 Problem Statement

Traditional examination of smart-meter data can require manual inspection of large volumes of consumption records. This makes it difficult to quickly identify unusual consumption patterns and understand their possible causes.

There is therefore a need for a software platform that can:

- organise and process smart-meter consumption data;
- identify unusual consumption behaviour;
- provide short-term load forecasting;
- present analytical results in an understandable form;
- provide explanations and contributing factors for flagged observations;
- support authorised personnel during investigation; and
- provide a unified dashboard for accessing the analytical results.

MeterIQ addresses these requirements by combining data analytics and visualisation in a single software-engineering system.

1.3 Project Objectives

The main objectives of MeterIQ are:

1. Understand and prepare smart-meter consumption datasets.
2. Perform exploratory data analysis to identify consumption patterns and unusual observations.
3. Develop suitable temporal and consumption-based features.
4. Develop an anomaly and electricity-theft-risk analysis module.
5. Develop a short-term load forecasting module.
6. Provide explainable analytical results rather than only binary decisions.
7. Develop an interactive dashboard for authorised users.
8. Integrate the analytical modules into a unified software system.

Methodology and Design

2.1 System Architecture

MeterIQ follows a modular architecture in which smart-meter data is first prepared and analysed before being passed to the analytical modules and dashboard.

The major processing stages are:

1. Smart-meter data input.
2. Data preprocessing.
3. Exploratory data analysis.
4. Feature engineering.
5. Anomaly and electricity-theft-risk analysis.
6. Short-term load forecasting.
7. Explainable dashboard.

The architecture allows the individual modules to be developed and tested independently before complete system integration.

2.1.1 High-Level Design

The high-level workflow of MeterIQ can be represented as:

**Smart-Meter Data → Data Preparation → Feature Engineering →
Analytics → Dashboard**

The analytics stage contains two major components:

- anomaly and electricity-theft-risk analysis; and
- short-term load forecasting.

The dashboard is intended to present the results of both components through an interactive interface.

2.1.2 System Design

The system is designed around four major actors or functional areas:

- smart-meter data source;
- data and analytics processing;
- authorised user; and
- dashboard and decision-support interface.

The system design diagrams described in the following sections provide different views of the same platform.

2.2 Software Engineering Design

The software engineering design of MeterIQ defines the interaction between users and the system, the flow of information through the platform, the underlying entities, and the overall operational workflow.

2.2.1 Use Case Diagram

The Use Case Diagram represents the interaction between the authorised user and the MeterIQ system.

The authorised user can access smart-meter information, analyse consumption, review anomaly and electricity-theft-risk indicators, inspect explanations, view forecasts, and use the dashboard for operational decision support.

2.2.2 Data Flow Diagram

The Data Flow Diagrams describe how smart-meter data moves through the MeterIQ system and how analytical results are delivered to users.

DFD Level 0

The Level 0 DFD provides a context-level representation of the MeterIQ system.

DFD Level 1

The Level 1 DFD decomposes the MeterIQ system into major processing stages, including data preparation, analytics, and presentation.

DFD Level 2

The Level 2 DFD provides a more detailed representation of the analytical processing performed within the system.

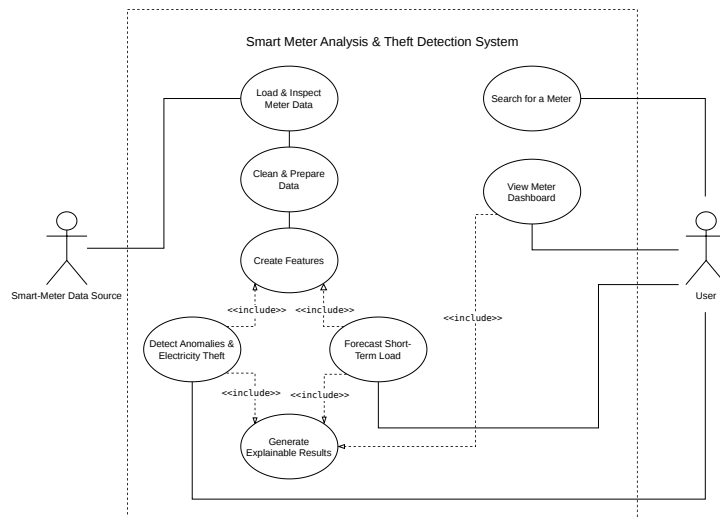


Figure 2.1: MeterIQ Use Case Diagram

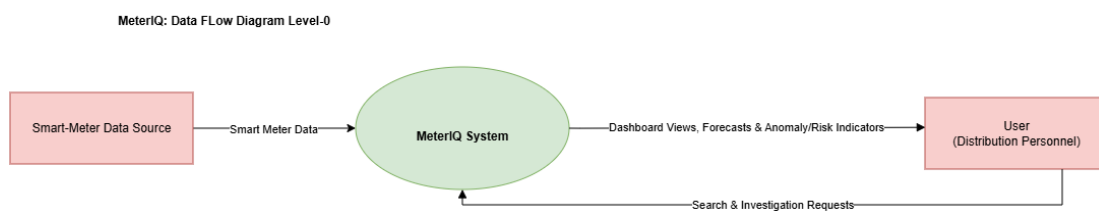


Figure 2.2: MeterIQ Data Flow Diagram Level 0

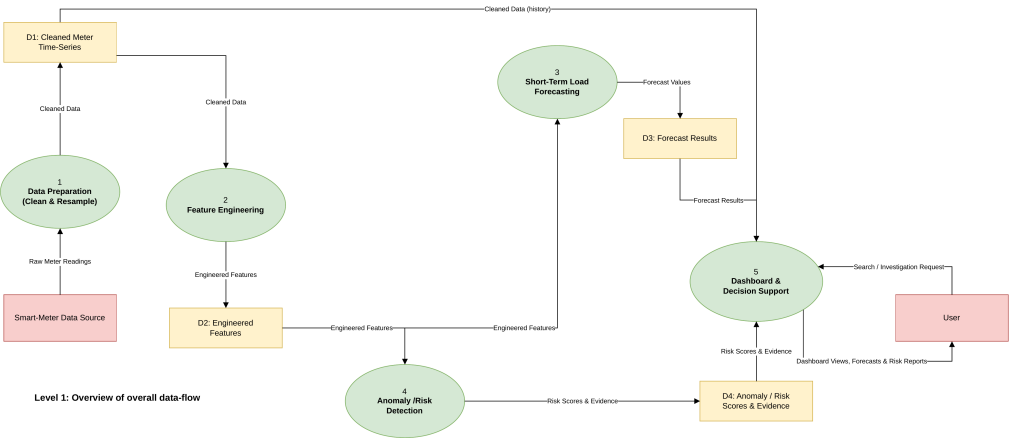


Figure 2.3: MeterIQ Data Flow Diagram Level 1

Level 2: Decomposition of Process 4 (Anomaly / Theft-Risk Detection)

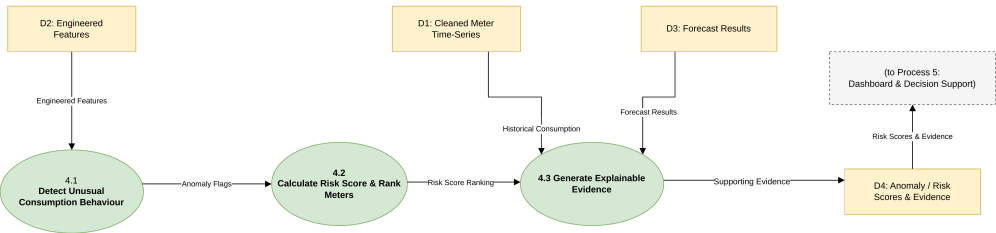


Figure 2.4: MeterIQ Data Flow Diagram Level 2

2.2.3 Entity Relationship Diagram

The Entity Relationship Diagram represents the major entities and their relationships within the MeterIQ system.

The database design supports smart-meter information, consumption records, analytical outputs, anomaly results, risk information, and investigation-related information.

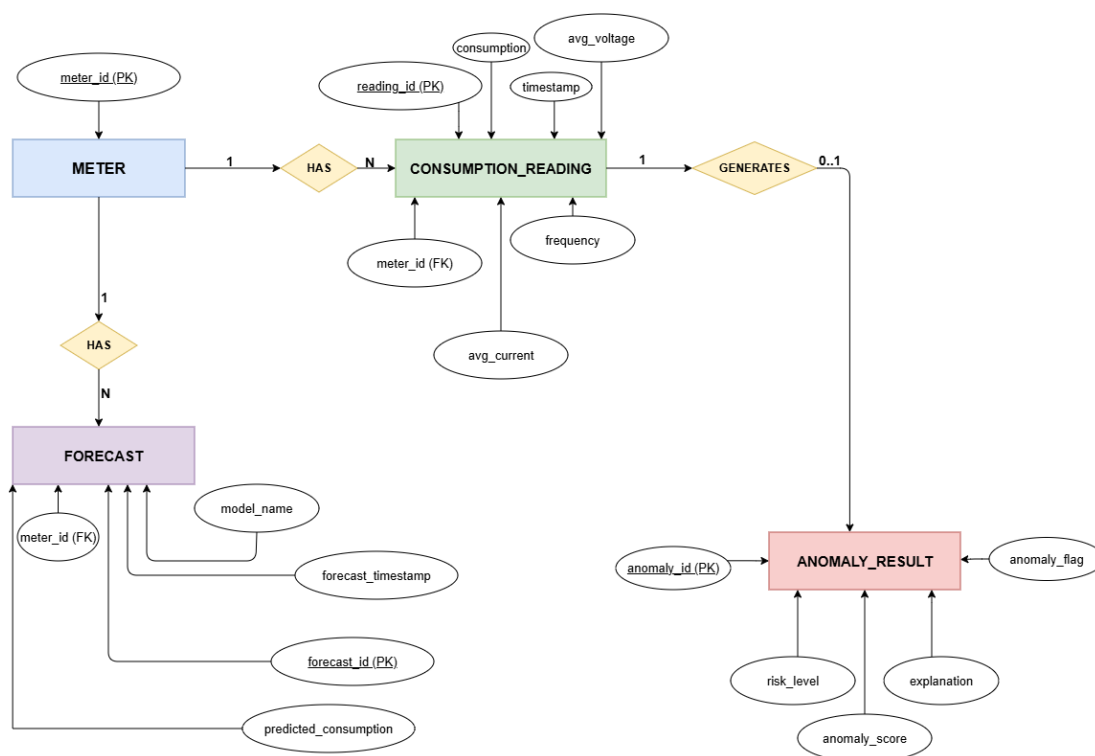


Figure 2.5: MeterIQ Entity Relationship Diagram

2.2.4 Activity Diagram

The Activity Diagram represents the sequence of activities involved in processing smart-meter information through the MeterIQ platform.

The general workflow starts with smart-meter data, followed by data preparation and feature generation. The prepared information is then used by the analytical modules, after which the results are presented to the authorised user.

2.3 Technical Stack

The project uses technologies appropriate for data analytics, machine learning, and interactive dashboard development.

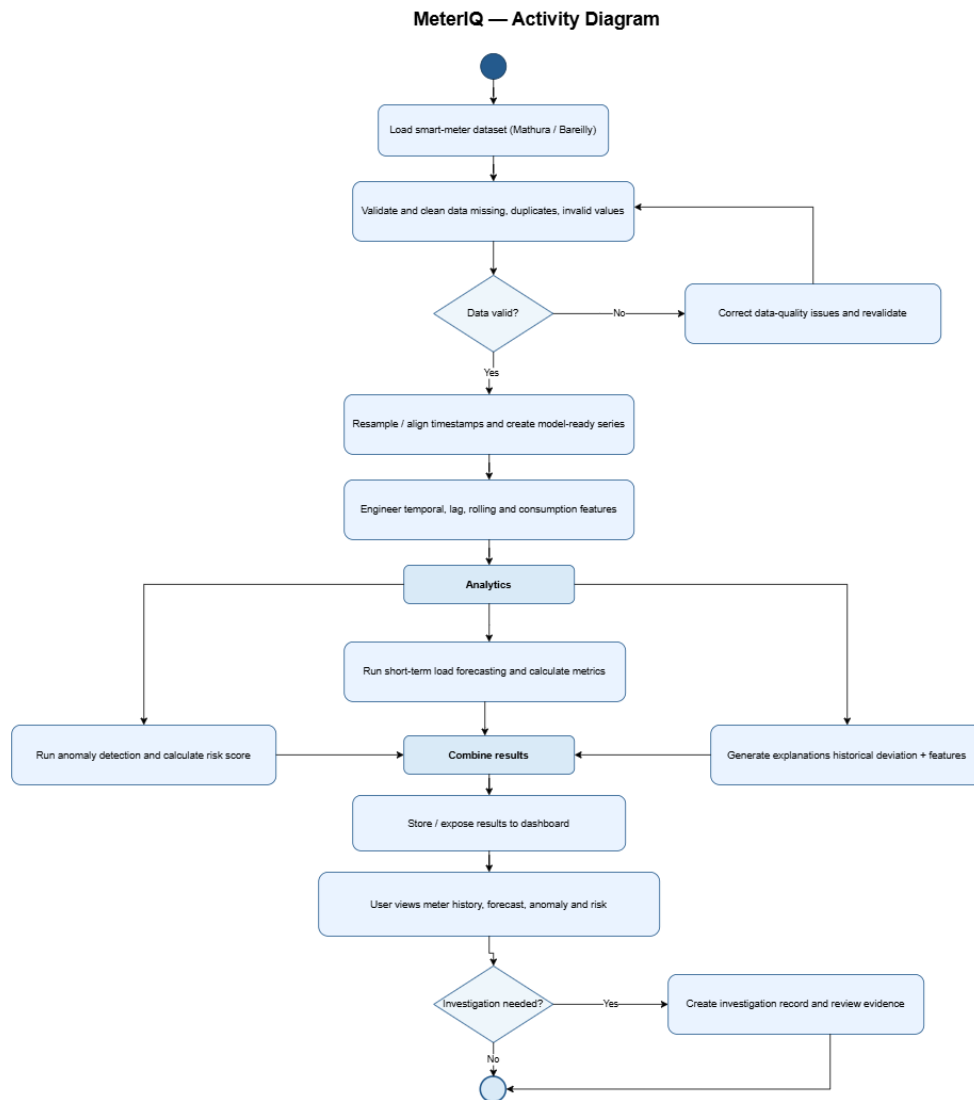


Figure 2.6: MeterIQ Activity Diagram

Table 2.1: MeterIQ Technical Stack

Area	Technology / Tool
Programming	Python, JavaScript
Data Analysis	Pandas, NumPy
Machine Learning	Scikit-learn
Visualisation	Matplotlib, Recharts
Frontend	React and Vite
Development	VS Code, Jupyter / Google Colab
Version Control	Git and GitHub
Dashboard	React-based interactive dashboard

2.4 Implementation Details

2.4.1 Data Preprocessing

The smart-meter data is first examined and prepared before analytical processing.

The preprocessing stage focuses on:

- checking the structure of the input datasets;
- handling date and time information;
- identifying duplicate observations;
- checking missing or inconsistent values;
- maintaining appropriate temporal ordering; and
- preparing the data for subsequent feature engineering.

The preprocessing stage is important because incorrect temporal ordering or data leakage can produce misleading analytical results.

2.4.2 Exploratory Data Analysis

Exploratory Data Analysis is used to understand the characteristics of smart-meter consumption data before model development.

The analysis focuses on:

- consumption distributions;
- temporal consumption patterns;
- hourly and daily behaviour;
- unusual consumption observations;
- variability in consumption; and
- relationships between temporal and consumption variables.

EDA helps establish an understanding of normal consumption behaviour before developing analytical models.

2.4.3 Feature Engineering

Temporal and lag-based features are created from the prepared smart-meter data.

The planned feature set includes:

- hour;
- day;

- month;
- weekday;
- lagged consumption;
- previous-day consumption; and
- previous-week consumption.

The same-hour-last-week concept is particularly relevant to the forecasting module because electricity demand can exhibit recurring weekly patterns.

Feature engineering is performed while maintaining chronological ordering to reduce the possibility of data leakage.

2.4.4 Module 1: Anomaly and Electricity-Theft-Risk Analysis

The first analytical module focuses on identifying unusual smart-meter consumption behaviour.

The purpose of this module is not to automatically declare a meter as committing electricity theft. Instead, unusual observations are converted into risk-oriented analytical information that can support investigation by authorised personnel.

The module is expected to consider:

- unusual consumption patterns;
- deviation from expected behaviour;
- temporal anomalies;
- relevant meter-level information;
- anomaly scores; and
- contributing factors that explain why an observation was flagged.

The final implementation and validation of the anomaly-detection component are part of the subsequent development stage.

2.4.5 Module 2: Short-Term Load Forecasting

The second analytical module focuses on short-term load forecasting.

The forecasting task is designed to estimate future electricity consumption for short horizons ranging from approximately 15 minutes to 24 hours.

A same-hour-last-week baseline is used as an important reference point. Forecasting performance will be evaluated using suitable error metrics such as Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE).

The evaluation will use a chronological holdout rather than a random split so that future observations do not influence training information.

2.4.6 Module 3: Explainable Dashboard

The dashboard provides a unified interface for accessing smart-meter analytics.

The dashboard is designed to provide:

- meter search and selection;
- consumption history;
- anomaly and risk information;
- contributing factors and explanations;
- short-term forecasts;
- visual analytics; and
- operational decision-support information.

The initial frontend implementation has been developed using React, Vite, and Recharts. At the current stage, the dashboard provides the interface foundation, while complete backend and machine-learning integration remains part of the next development stage.

2.4.7 Backend and Database Development

The backend and persistent database layer are planned for the next stage of development.

The database will provide persistent storage for relevant meter information, consumption records, analytical outputs, risk information, forecast outputs, and investigation-related information.

The backend and database will subsequently be integrated with the existing frontend and machine-learning modules.

Results and Evaluation

3.1 Current Project Status

The current mid-semester development has established the software-engineering and data-analysis foundation of the MeterIQ platform.

The following activities have been completed or substantially developed:

1. Project planning and Gantt chart preparation.
2. Requirements and system-level design.
3. Use Case Diagram.
4. Data Flow Diagrams.
5. Exploratory Data Analysis.
6. Data preprocessing.
7. Feature engineering.
8. Prepared dataset validation.
9. Initial dashboard development.

The remaining development activities include complete machine-learning implementation, forecasting validation, backend development, persistent database integration, complete system integration, and final testing.

3.2 Testing Methodology

Testing will be performed at multiple levels to verify the correctness and reliability of the MeterIQ system.

3.2.1 Data Testing

Data testing will verify:

- input structure;
- missing values;
- duplicate records;
- temporal ordering;

- feature consistency; and
- processed dataset validity.

3.2.2 Model Testing

The anomaly and forecasting modules will be evaluated using appropriate datasets and evaluation procedures.

For anomaly analysis, the evaluation will focus on whether unusual behaviour can be identified consistently and whether the generated explanations correspond to the underlying features.

For forecasting, chronological validation will be used to measure forecasting errors without introducing future information into the training stage.

3.2.3 Interface Testing

The dashboard will be tested for:

- navigation;
- meter selection;
- visualisation rendering;
- display of analytical results;
- explanation presentation; and
- usability of the main workflow.

3.2.4 Integration Testing

Integration testing will verify the complete flow:

Data → Processing → Analytics → Backend → Dashboard

3.3 Performance Metrics

The final evaluation will use metrics appropriate to each analytical component.

For forecasting, the primary metrics will include:

- Mean Absolute Error (MAE);
- Mean Absolute Percentage Error (MAPE); and
- comparison against the same-hour-last-week baseline.

For anomaly analysis, evaluation will depend on the availability and quality of appropriate validation information. The final metrics will therefore be reported after the anomaly-detection implementation and validation have been completed.

No final model accuracy values are reported at the current stage because the machine-learning modules are still under development.

3.4 Analysis

The current implementation demonstrates the foundation required for an integrated smart-meter analytics platform.

The project has progressed from data understanding and preparation toward the development of analytical and presentation components.

An important design principle is explainability. Instead of presenting only a binary decision, the system is intended to present risk-oriented information together with contributing factors so that authorised users can perform further investigation.

The final analysis will be completed after the anomaly-detection, forecasting, backend, and integration stages have been implemented and validated.

Conclusion

4.1 Summary of Work

MeterIQ is being developed as an explainable smart-meter analytics platform for supporting distribution-grid monitoring and analysis.

The project combines smart-meter data preparation, exploratory data analysis, feature engineering, anomaly and electricity-theft-risk analysis, short-term load forecasting, and an interactive dashboard within a unified software system.

At the current stage, the major software-engineering diagrams, data exploration, pre-processing, feature engineering, prepared dataset validation, project planning, and initial dashboard implementation have been completed.

4.2 Key Findings

The current development has established the following:

- **Unified Analytics:** Different smart-meter analytical functions can be organised within a single software platform.
- **Explainability:** Anomaly analysis can be presented using risk-oriented information and contributing factors rather than only a binary decision.
- **Human-in-the-Loop:** Electricity-theft-risk signals are intended to support further investigation by authorised personnel rather than automatically determining theft.
- **Temporal Analysis:** Smart-meter data contains temporal patterns that are useful for both anomaly analysis and forecasting.
- **Dashboard Integration:** An interactive dashboard provides a common interface through which different analytical outputs can eventually be accessed.

4.3 Future Work and Recommendations

The planned future work includes:

1. Complete implementation and validation of the anomaly-detection module.
2. Develop and evaluate short-term load forecasting models.
3. Compare forecasting models against the same-hour-last-week baseline.

4. Integrate the machine-learning modules with the dashboard.
5. Implement backend services.
6. Implement persistent database storage.
7. Connect processed smart-meter datasets to the complete system.
8. Complete end-to-end system testing.
9. Improve explanation and investigation workflows.
10. Evaluate the final system using appropriate performance metrics.
11. Further improve the analytical modules based on evaluation results.

4.4 Final Remarks

MeterIQ aims to provide an integrated and explainable approach to smart-meter analytics.

The system is designed to assist authorised operators by highlighting unusual patterns, providing forecasting information, and presenting analytical evidence through an interactive interface.

The current mid-semester implementation establishes the foundation for the remaining machine-learning, backend, database, integration, and testing stages.